

Database Indexing - Lecture Notes

A database index is an auxiliary data structure that lets the engine find matching rows without scanning the whole table, which greatly speeds up reads.

B+ tree indexes keep keys sorted and linked at the leaf level, so they support both point lookups and range queries efficiently.

A clustered index defines the physical order of rows on disk; a table can have at most one. Secondary indexes store the key plus a pointer to the row.

The trade-off of adding an index: extra storage, and slower writes, because every INSERT, UPDATE and DELETE must also maintain the index.